Do Now · retrieval
Floating-point subtraction

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why must floating-point exponents match before the mantissas are added directly?

Q2

Convert 01100000 1110 to denary. Use an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Q3

Convert $6D_{16}$ to eight-bit binary.

Q4

Use 8-bit two’s complement to calculate $18 - 29$. Show how subtraction becomes addition.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Align before subtracting — Choose a common exponent and use arithmetic right shifts to align the mantissas. Keep any low-order guard bits needed for an exact working result.

Core idea 2

Negate the second mantissa — At the chosen working width, invert every bit of B’s aligned mantissa and add 1. — CODE:
A − B
= A + (−B)

Core idea 3

Add the aligned values — Sign-extend both operands to the same working width, add them, and interpret the signed result using the common exponent.

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Handle cancellation, zero and range — A small difference may need left shifts and a reduced exponent to become normalised. Equal operands produce zero.

Vocabulary · use the correct technical terms

Negation — Changing the sign of a value; in two’s complement, invert and add 1.

Cancellation — When subtracting similar values produces a much smaller result.

Guard bits — Extra low-order working bits retained during calculation.

Arithmetic right shift — A right shift that repeats the sign bit.

Independent skill check · apply the lesson

Calculate 5 - (-1.5) with a 6-bit two's-complement mantissa and 4-bit exponent. Show align, negate, add and normalise.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Subtract 1.5 from 5

Follow the worked method

1. Use common exponent 3 — Align both mantissas using sign-preserving right shifts. — CODE:
A: 0010100
B: 0000110

2. Calculate A + (−B) — CODE:
0010100
+ 1111010
= 0001110 — The working difference is $14/32 \times 2^3$.

3. Shift left and reduce the exponent — CODE:
0001110×2^3
$\rightarrow 011100 \times 2^2$

Final: 011100 0010 — Check: 5 − 1.5 = 3.5 .

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · solo guided practice

Work ALONE. Try first; read the clues one at a time if you get stuck.

Complete the steps below. Check your answer only on page 8.

YOUR TASK

Build the answer

CLUES · uncover one at a time

Clue 1: Rewrite 1.5 - 5 as 1.5 + (-5).

Clue 2: Align to the same exponent BEFORE negating the second mantissa.

Clue 3: Keep an extra sign bit in the working sum; then normalise.

BUILD YOUR ANSWER · in BLACK PEN

Step 1: Align the mantissas.

Step 2: Negate the second and add.

Step 3: Normalise the signed result and check its value.

CHECK BEFORE TURNING THE PAGE

Have you shown every step? The worked self-check is at the END on page 8.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Question 1 · ADAPTED OCR EXAM PRACTICE · [6]
Calculate 010010 0010 minus 010010 0100 . Show exact alignment, negation, addition and an answer in normalised floating-point form using a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent.
Adapted from OCR June 2017 Q6(d): operand order reversed; classroom marks.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Question 4 · ORIGINAL EXAM-STYLE EXPLANATIONS · [4]
State the stored result when a value is subtracted from itself, using a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. [1]
Explain why normalising a small difference can require a lower exponent. [2]
Explain why a nonzero low-order bit must be retained during alignment where possible. [1]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

When subtracting B from A, only B's mantissa is negated before addition.

Use key vocabulary from memory

1. Explain Negation in your own words.

2. Explain Cancellation in your own words.

3. Explain Guard bits in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

END SELF-CHECK · only after you have attempted the work

PAGE 5 · Guided We Do solution:

Aligned at exponent 3: 0000110 + 1101100 = 1110010; normalised 100100 0010 = -3.5.

PAGE 3 · Independent skill check:

5 - (-1.5) = +6.5. Common exponent 3; aligned mantissas 0010100 and 1111010; negate second to 0000110; sum 0011010; normalised result 011010 0011.

Correct any errors in GREEN PEN. Write one improvement.

YOU DO ANSWERS: rhsccs.org/lessons/h446/lesson-007/answers.html